

THE GEOMETRIC BOUNCE HYPOTHESIS:

Recursive Physical Homeostasis from Galactic Seeds to Planetary Purification

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Abstract

We present a unified cosmological model based on the Axiomatic Physical Homeostasis (APH) framework. We propose that the Universe is a recursive geometric structure formed by the filtration of the Cayley-Dickson algebras. We trace the lineage of structure formation from the Primordial Sedenion Seeds that anchored the Milky Way to the *Geometric Bounce* event that purified the Earth-Moon system. We resolve the Isotopic Crisis of lunar formation by identifying the Moon as the ejected Associative Residue of a planetary-scale vacuum phase transition. Furthermore, we demonstrate that the *Great Filter* of the Fermi Paradox is identifiable as the Geometric Bounce itself: a necessary but perilous stiffness excursion required to stabilize a planetary reactor for biological observability. Finally, we map the Cosmic Microwave Background (CMB) anomalies and the void structure to the topology of excluded Sedenion zero divisors, predicting a specific gravitational wave signature at 1.8 GHz.

1 Introduction: The Fractal Filtration

Standard cosmology treats structure formation as a bottom-up hierarchical assembly of cold dark matter. In contrast, the APH framework posits a top-down *Geometric Filtration* driven by the relaxation of vacuum stiffness $\beta(t)$. The universe began in a state of Sedenion Chaos ($\beta \approx 0$), characterized by Rule 30-class cellular automaton behavior. The transition to Octonionic Order ($\beta \approx 1.91$) did not occur uniformly but fractured the vacuum into distinct strata:

1. **The Bulk (Voids):** Regions of excluded Sedenion zero divisors.
2. **The Filament (Matter):** Stable Octonionic geometry allowing for associative physics.
3. **The Interface (Seeds):** High-density topological defects that served as the nuclei for galaxies.

This paper explores how this filtration process repeats recursively at the galactic and planetary scales.

2 Galactic Genesis: The Sedenion Seed

Standard models struggle to explain the rapid formation of supermassive black holes (SMBHs) at high redshifts ($z > 7$). APH resolves this by inverting the hierarchy: the black hole formed first.

2.1 Fractal Dust and Direct Collapse

The primordial phase transition produced *Sedenion Dust*—topological defects with a fractal Hausdorff dimension of $D_F \approx 2$ (Cosmic Sheets). These defects acted as Direct Collapse Seeds with initial masses of $10^4 - 10^6 M_\odot$, bypassing the stellar-mass bottleneck. The Milky Way formed around one such seed (Sagittarius A*). This core is not a singularity but a Sedenion Star governed by a super-stiff equation of state:

$$P = K\rho^\pi, \quad \Gamma_S \equiv \pi \approx 3.14159 \quad (1)$$

This stiffness allows the core to support mass without collapsing to zero radius, stabilized by internal algebraic convection.

2.2 The Halo as Geometric Surface Tension

The galactic halo is not composed of particulate dark matter but is the manifestation of the elastic potential energy of the G_2 domain wall wrapping the galaxy. Using the APH Geometric Surface Virial Theorem, the flat rotation curves are derived from the surface tension σ_{wall} of the interface between the Octonionic galaxy and the Sedenion void:

$$v(r) = \sqrt{\frac{GM_{bar}}{r} + v_0^2} \quad (2)$$

where v_0 is the characteristic velocity of the domain wall tension.

3 The Geometric Bounce: Planetary Purification

The solar system is a recursive instance of the galactic pattern. The planets act as *Stiffness Nodes* designed to minimize local Associator Hazard.

3.1 The Earth as a Stiffness Reactor

The Proto-Earth functioned as a Geometric Stiffness Reactor (GSR), concentrating high-hazard elements (Iron/Nickel) in the core to induce vacuum stiffness $\beta \gg 1$. Under APH, the Moon's formation is the result of a control failure in this reactor—a *Geometric Bounce*. A transient excursion of the stiffness parameter generated a repulsive pressure P_{geom} exceeding the gravitational binding energy:

$$\int P_{geom} dV > \frac{3GM_\oplus^2}{5R} \quad (3)$$

3.2 Solving the Isotopic Crisis

This endogenic model resolves the fine-tuning problems of the Giant Impact Hypothesis:

- **Isotopes:** The Moon is formed from Proto-Earth mantle; thus, $\delta^{17}O$ and $\delta^{50}Ti$ signatures are identical by definition.
- **Iron Depletion:** The high-hazard Iron core was retained by the reactor (Earth) to maintain stiffness, while the low-hazard silicate *slag* was ejected (Moon) via Associator Sorting.
- **Energy Source:** The separation energy $E \approx 3 \times 10^{30}$ J was supplied by the Latent Heat of the vacuum phase transition, not kinetic impact.

4 Cosmic Voids and the CMB Structure

The large-scale structure of the universe reflects the geometry of the *swampland* rejected by the APH filtration.

4.1 Voids as Sedenion Fossils

Cosmic voids are not empty; they are regions of the Sedenion Bulk ($\mathcal{S}$) where the Associator Anomaly persists. They are causally disconnected regions of high stiffness ($\Gamma = \pi$) that exert the repulsive pressure mimicking Dark Energy. The expansion of the universe is driven by the relaxation of this global stiffness from $\pi \rightarrow 2.91$.

4.2 The Sedenion Scream (1.8 GHz)

The APH model predicts that the phase transition left a spectral fingerprint. The Hydrogen hyperfine line ($\nu_H \approx 1.42$ GHz) acts as the anchor for the vacuum lattice. The decay of S^3 zero-divisor fibers generates a gravitational wave peak at:

$$f_{peak} \approx \frac{\nu_H}{\sqrt{2}} \cdot \frac{3}{2} \zeta(3) \approx 1.81 \text{ GHz} \quad (4)$$

This *Sedenion Scream* is a falsifiable prediction detectable by Axion Haloscopes via the Inverse Gertsenshtein Effect.

5 The Fermi Paradox: The Bounce as the Great Filter

The APH framework offers a purely geometric resolution to the Fermi Paradox. The *Great Filter* is not nuclear war or biological improbability, but the *Geometric Bounce* itself.

5.1 The Anthropogeometric Principle

Axiom 2 of APH (Observability) requires a unitary, stable metric for an observer to exist.

- A planet must possess a differentiated core to generate a magnetic field (shielding).
- However, a high-mass core generates excessive Associator Hazard ($\mathcal{A}(Z)$).
- To stabilize, the planet *must* undergo a Geometric Bounce to eject the associative residue (forming a Moon) and stabilize the core stiffness.

The Hazard: This phase transition is violent. If the excursion is too weak, the planet remains unstable (Venus-like, no moon, chaotic spin). If too strong, the planet is obliterated (Asteroid Belt). Only those rare *Goldilocks* reactors that successfully execute a controlled Bounce—retaining the core while ejecting the mantle—can support long-term biological evolution. We are alone because we are the survivors of a planetary-scale vacuum explosion.

6 Conclusion

From the galactic seeds of the Sedenion era to the recursive purification of the Earth, the universe is a machine striving for Homeostasis. The Moon is not an accident; it is the slag of our survival. The Milky Way is not a random spiral; it is a crystallized node of Order. We have unified the Dark Sector, the Isotopic Crisis, and the Silence of the Stars into a single, computable geometric history.

References

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